



Gabriel Peytral Borja

M2 Applied Mathematics & Statistics · Computational and Mathematical Biology

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Master's student in Applied Mathematics and Statistics (Computational & Mathematical Biology track) at Aix-Marseille Université. I work on probabilistic and state-space models of physiological signals: cardiac electrical activity in my bachelor's thesis, and neural fiber-photometry recordings at the Abante Lab in Barcelona. At ease across mathematics, statistics, scientific computing and the life sciences. Seeking a six-month end-of-studies research internship in statistical modelling and machine learning for health, from February 2027, as a step towards doctoral research.

RESEARCH EXPERIENCE

Research Intern — Abante Lab

05/2026 – 07/2026

Universitat de Barcelona · PI: Jordi Abante (UBNeuro / IDIBAPS) Barcelona, Spain

- Developed **photosvi**, a composable Python package of generative models for **fiber-photometry signals**, used for drug-response profiling of neural recordings; refactored one-off research scripts into a single config-driven architecture.
- Implemented **variational inference** (Pyro, SVI) with **log-space forward-backward and Viterbi decoders**, validated against brute-force oracles.
- Built the **benchmark** comparing five inference variants for **phasic-event detection and drug-effect recovery** against simulated ground truth; the selected variant became the lab's baseline.
- Results feed a **manuscript in preparation**.

Bachelor's Thesis — Discrete Fuzzy Takagi-Sugeno Model of Cardiac Electrical Activity

09/2024 – 03/2025

UNAM, Artificial Intelligence Department · funded by CONACYT Querétaro, Mexico

- Designed and implemented a discrete-time **Takagi-Sugeno fuzzy model** reproducing the electrical activity of the heart, generating the ECG live from the model.
- Achieved a **~400% speed-up** in generating cardiac rhythms, with approximation errors in the **millisecond range or below**, fast enough to run on a microcontroller.

EDUCATION

MSc Applied Mathematics, Statistics

09/2025 – present

Aix-Marseille Université · track: Computational & Mathematical Biology (CMB) Marseille, France

- TIGER** master's excellence scholarship.

Bachelor's Degree in Technology (Licenciatura en Tecnología)

09/2020 – 03/2025

National Autonomous University of Mexico (UNAM) Querétaro, Mexico

- GPA 9.34/10**, graduated with honours, top ~5% of the cohort.
- Interdisciplinary programme structured around research training; electives focused on computer science and mathematics.

Erasmus exchange

02/2024 – 07/2024

University of Groningen · full UNAM excellence mobility scholarship Groningen, Netherlands

- Courses: Numerical Mathematics · Neural Networks · Operating Systems.
- Team project: **CNN reconstruction** of Gaussian-blurred image regions (OpenImages).

METHODS

Bayesian & variational inference

SVI, ELBO

State-space models

HMMs, Deep Markov Models; forward-backward, Viterbi

Statistical modelling of physiological signals

Numerical methods & scientific computing

Machine learning, neural networks

COMPUTING

Python

Pyro, PyTorch, NumPy/SciPy: probabilistic modelling & deep learning

R

Statistical analysis

C++, CUDA, OpenMP

Parallel and GPU programming

Java, Spring Boot

OOP, enterprise applications

Linux/Unix, Bash, Git

Remote servers, reproducible workflows

■ UNDERGRADUATE RESEARCH STAYS

Research stay — Institute of Mathematics (IMATE) 01/2023 – 05/2023

UNAM · Dr. Guillermo Ramírez Santiago

- Mathematical modelling of **active-matter systems** with fluid dynamics and tensor calculus, applied to bacterial aggregation into **biofilms**.

Research stay — Nanobiophotonics Laboratory 01/2022 – 05/2022

CFATA, UNAM · Dr. Luz López Marín

Querétaro, Mexico

- Wet-lab work on a **COVID-19 aptamer biosensor**, then **dendritic-cell** culture and activation under **shockwaves** (a cancer-immunotherapy angle); protocol design and reporting.

■ PROFESSIONAL EXPERIENCE

Junior Software Developer

Intellect Design Arena Ltd

03/2025 – 07/2025

Mexico City, Mexico

- Wrote functional and technical **user stories** and developed **Spring Boot** utilities for enterprise banking applications.

■ SELECTED PROJECTS

Heat-transfer simulation & wildfire propagation

UNAM · team project, later with Dr. Solís

2022 – 2023

Querétaro, Mexico

- Numerical solution of the **heat-transfer PDE** from a photograph plus a material specification (OpenCV), extended to **wildfire spread over 3D terrain** with **simulated annealing**. **Best project of the year**; led to an invitation to collaborate with SEMARNAT.

CNN-LSTM ground-humidity forecasting

UNAM

2024

Querétaro, Mexico

- **CNN-LSTM** architecture for long-horizon humidity prediction on large global gridded datasets (Python, TensorFlow).

Real-time parallel Sobel edge detection

UNAM · image processing & parallel computing electives

2023

Querétaro, Mexico

- Filters, noise reduction and segmentation from scratch, culminating in a **real-time parallel Sobel detector** (C++, OpenMP, CUDA).

Air pollution & COVID-19 spread

UNAM · with a peer

2021

Mexico City, Mexico

- Statistical model relating particulate-matter pollution to COVID-19 propagation; explained **~8% of contagions** in the target region.

Cove — a concept-graph language-learning app

Side project, in development

2026 –

- A learner-built graph of concepts across several languages, a rule-based grammatical core, and a generator calibrated to what the graph holds (Flutter, Dart, SQLite, spaCy).

■ LANGUAGES

Spanish

Native

English

C1 · Cambridge C1, TOEFL iBT 105

French

C1 · TCF

German

A2

■ ONLINE

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